

Shivani Sanaboina

Ai Engineer

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Professional Summary

AI/ML Engineer with 3+ years of experience designing and deploying machine learning, deep learning, and Generative AI solutions across financial services and enterprise environments. Skilled in Transformer-based NLP, LLM fine-tuning, and Retrieval-Augmented Generation (RAG) with expertise in multi-agent AI frameworks, cloud deployment on AWS, and MLOps practices. Proven track record in building production-grade fraud detection, financial risk prediction, and compliance automation systems using PyTorch, TensorFlow, LangChain, and Hugging Face Transformers.

Technical Skills:

- **Programming & Data:** Python, SQL, PySpark, Pandas; data wrangling, feature engineering, statistical analysis, ETL pipelines
- **Machine Learning & Deep Learning:** Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Keras; CNNs, RNNs, LSTMs, Transformers; supervised/unsupervised modeling, hyperparameter tuning, A/B testing
- **NLP & Generative AI:** BERT, GPT, LLaMA, Hugging Face Transformers, LangChain, LlamaIndex, RAG, prompt engineering, RLHF, multi-agent frameworks (LangGraph, AutoGen, Semantic Kernel)
- **Cloud & Deployment:** AWS (SageMaker, Bedrock, Lambda, S3, API Gateway), Docker, Kubernetes, Flask APIs, CI/CD pipelines
- **MLOps & Monitoring:** MLflow, AWS CloudWatch, drift detection, model versioning, governance, automated retraining
- **Visualization & Databases:** Tableau, Matplotlib, Seaborn, Power BI; PostgreSQL, Pinecone, Chroma, MySQL

Work Experience:

JP Morgan & Chase, NY | AI Engineer

Mar 2025 – Present

- Developed Generative AI, Transformer-based NLP, and Machine Learning models for financial risk prediction, fraud detection, and compliance monitoring using Python, PyTorch, TensorFlow, Scikit-learn, and Hugging Face Transformers.
- Built Retrieval-Augmented Generation (RAG) systems with LangChain, LlamaIndex, Pinecone, and Chroma, integrating transformer embeddings and hybrid search for knowledge retrieval from large-scale financial datasets.
- Designed multi-agent AI workflows leveraging LangGraph, AutoGen, and Semantic Kernel with transformer-based reasoning agents to enable intelligent decision-making, automated report generation, and anomaly detection across financial datasets.
- Conducted data wrangling, feature engineering, and statistical analysis on large transactional datasets using PySpark, SQL, and Pandas, improving Transformer model accuracy and business insights.
- Leveraged AWS (SageMaker, Bedrock, Lambda, S3) for training and deployment of Transformer-based LLMs and agentic AI systems, ensuring compliance with enterprise security standards and scalable financial production pipelines.
- Applied prompt engineering and reinforcement learning with human feedback (RLHF) to optimize LLM contextual accuracy, reduce hallucinations, and enhance Transformer-based financial document interpretation and classification tasks.
- Partnered with quant analysts and business stakeholders to conduct predictive modeling, hypothesis testing, and A/B experiments that influenced strategic decisions using Transformer models and multi-agent frameworks.
- Delivered insights via Tableau, Matplotlib, and Seaborn dashboards, improving leadership transparency into AI agent workflows, Transformer predictions, and model interpretability for regulatory, compliance, and risk assessment initiatives.

Accenture | Machine Learning Engineer | Hyderabad, India

May 2022 – Jun 2024

- Devised deep learning models including CNN and RNN architectures using TensorFlow and PyTorch for automated document classification and fraud transaction detection, increasing operational throughput and reducing manual review efforts.
- Optimized and fine-tuned BERT-base and GPT-2 models via Hugging Face Transformers for customer sentiment analysis and intent detection across support tickets, improving NLP classification accuracy by 20% and reducing customer escalation times.
- Formed reusable PyTorch and TensorFlow training pipelines with automated hyperparameter tuning and batch inference, reducing model training time by 60% and ensuring reproducibility and scalability across deployments.
- Formulated and deployed real-time anomaly detection services using LSTM networks and OpenCV-powered transaction receipt parsing via Flask APIs, reducing fraud monitoring efforts by 55% and strengthening real-time compliance controls.
- Transformed legacy Scikit-learn and XGBoost models to AWS SageMaker using containerized deployments with Docker, establishing API Gateway and Lambda endpoints, reducing deployment time by 50% and enabling seamless auto-scaling.
- Orchestrated model evaluation, logging, and drift detection with MLflow and AWS CloudWatch, enhancing governance, reducing production incidents by 40%, and ensuring continuous model performance and reliability across deployed models.
- Crafted and maintained Tableau dashboards integrated with PostgreSQL queries, prediction confidence intervals, and model impact KPIs, boosting stakeholder decision-making efficiency and improved data-driven strategy adoption.

Education:

- **Masters in management information systems** from Robert Morris University, Moon Township, PA, USA.
- **Bachelors in computer science and engineering** from CMR Institute of Technology, Hyderabad, India.